

CSCI 5541 NLP HW1

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1 Introduction

The goal of this assignment is to use n-gram language models to write a program that will be able to classify authors based on a set of training text. This report shows the result in different n and sets different configurations to improve performance. As assignment asks, the report will include the following: What information is in your model? What method of smoothing did you use ? What else did you include to improve performance? Which of your different configurations performed the best?

2 Methodology

Our model implements a generative n-gram approach. We utilize the o200k_base tokenizer from *tiktoken* to convert raw text into sequences of tokens.

2.1 Model Details

The model tracks counts for all sequences from unigrams up to the specified n . We implemented a global vocabulary approach where the vocabulary size $|V|$ is determined by the union of all training sets. This ensures that the probability distributions for different authors are normalized over the same space. And try to tuning hyperparameter n and k to improve the result of the performance.

2.2 Add-k Smoothing

We applied Add-k Smoothing to deal with the 0 probability problem of n-gram data. The probability of a token w_n given its context w_{n-1} is calculated as:

$$P(w_n|w_{n-1}) = \frac{\text{count}(w_n, w_{n-1}) + k}{\text{count}(w_{n-1}) + k \cdot |V|}$$

In our experiments, choose different k .

2.3 interpolation

And I applied the interpolation try to improve the model which calculate as and set default as trigram

LM:

$$P(w_3 | w_1, w_2) = \lambda_1 P(w_3 | w_1, w_2) + \lambda_2 P(w_3 | w_2) + \lambda_3 P(w_3)$$

$$\sum_i \lambda_i = 1$$

3 Experimental Results

3.1 test inside

First, we compared the performance of our classifier at different values of n with Laplace Smoothing. when $n = 1$:

Author	Accuracy (%)
Austen	82.0
Dickens	69.9
Tolstoy	72.7
Wilde	69.1
Overall (total_correct / total_sentences)	73.41

when $n = 3$:

Author	Accuracy (%)
Austen	67.7
Dickens	64.2
Tolstoy	61.2
Wilde	60.7
Overall (total_correct / total_sentences)	63.45

when $n = 10$:

Author	Accuracy (%)
Austen	53.0
Dickens	29.8
Tolstoy	17.1
Wilde	18.0
Overall (total_correct / total_sentences)	29.42

Second, we compared the performance of our classifier at different values of smoothing k with trigram model.

when $k = 0.5$:

Author	Accuracy (%)
Austen	68.3
Dickens	64.1
Tolstoy	62.1
Wilde	61.6
Overall (total_correct / total_sentences)	64.00

when $k = 0.01$:

Author	Accuracy (%)
Austen	66.2
Dickens	62.4
Tolstoy	61.7
Wilde	61.1
Overall (total_correct / total_sentences)	62.86

when $k = 10$:

Author	Accuracy (%)
Austen	61.7
Dickens	60.6
Tolstoy	55.4
Wilde	53.5
Overall (total_correct / total_sentences)	57.79

This section compare the coefficient of interpolation. As I show above using less n have the better performance. Hence, I will give more weight at the unigram probability. Which compare with the equally weight in condition of the trigram model with $k = 0.5$.

Case 1 : $\lambda_{unigram} = 0.6$; $\lambda_2 = 0.3$; $\lambda_3 = 0.1$

Author	Accuracy (%)
Austen	84.9
Dickens	71.3
Tolstoy	74.3
Wilde	69.9
Overall (total_correct / total_sentences)	75.08

Case 2 : $\lambda_{unigram} = \frac{1}{3}$; $\lambda_2 = \frac{1}{3}$; $\lambda_3 = \frac{1}{3}$

Author	Accuracy (%)
Austen	84.0
Dickens	71.0
Tolstoy	74.0
Wilde	69.8
Overall (total_correct / total_sentences)	74.69

3.2 test outside

This section shows the results on the held-out test data, obtained by running: `python3`

`classifier.py` `authorlist` `-test` `[filename]`. I will compare two LM, both are trigram models difference between them is using interpolation or not. (with $k = 0.5$; $\lambda_{unigram} = 0.6$; $\lambda_2 = 0.3$; $\lambda_3 = 0.1$)

First, I choose the first chapter of "Persuasion", which is not contain in the training data mention at homework website. The result are following:

Table 1: Classification distribution on Austen's sentences

Interpolation	OFF (%)	ON (%)
Austen	75.0	83.3
Dickens	8.3	0.0
Tolstoy	8.3	4.2
Wilde	8.3	12.5
Total sentences	24	24

Second, using the first chapter of "bleak house" as test sample, which wrote by Dickens

Table 2: Classification distribution on Dickens's sentences

Interpolation	OFF (%)	ON (%)
Dickens	50.0	50.0
Austen	25.0	14.3
Tolstoy	14.3	17.9
Wilde	10.7	17.9
Total sentences	28	28

Third, using the first chapter of "A confession" as test sample, which wrote by Tolstoy.

Table 3: Classification distribution on Tolstoy's sentences

Interpolation	OFF (%)	ON (%)
Tolstoy	46.2	23.1
Dickens	46.2	30.8
Austen	7.7	15.4
Wilde	0.0	30.8
Total sentences	13	13

Finally, using the first chapter of "The Picture of Dorian Gray" as test sample, which wrote by Wilde.

Table 4: Classification distribution on Wilde's sentences

Interpolation	OFF (%)	ON (%)
Wilde	96.3	92.6
Tolstoy	3.7	4.9
Austen	0.0	2.5
Dickens	0.0	0.0
Total sentences	81	81

4 Summary

I do ablation test in this assignment, only change 1 condition and remain others same. All in all, The best model is using Add 0.5 smoothing and which $n = 3$, and then do the linear interpolation in training dataset. By the way, the higher order probability is given by

$$P(w_n | w_{n-1}, w_{n-2}) = \frac{\text{count}(w_n, w_{n-1}, w_{n-2}) + k}{\text{count}(w_{n-1}) + k \cdot |V|}$$

However, in the phase of testing. The model seems overfit, which only have good performance in Austen's book.

5 Reference

References

- [1] Dan Jurafsky and James H. Martin, *Speech and Language Processing*, 3rd edition
- [2] Leo Tolstoy, *A Confession*, translation by Louise and Aylmer Maude
- [3] Oscar Wilde, *The Picture of Dorian Gray*
- [4] Charles Dickens, *Bleak House*
- [5] Jane Austen, *Persuasion*